

KIU Assignment Writer

Bugs ki Mukammal Report — Roman Urdu

Code review ke baad | Tareekh: 04 June 2026

1. Taaruf

Yeh report aap ke "Write Assignment" tool ke poore code ko parhne ke baad banayi gayi hai. Isme woh saare files shaamil hain jo assignment likhne, page banane, aur PDF export karne ka kaam karte hain:

- assignment_geometry.dart — page ki naap-tol (chaurai, margins, font size, line height, ek page mein kitni lines).
- assignment_html_service.dart — editor ka HTML + JavaScript, aur PDF ka HTML.
- web_assignment_editor_screen.dart — Flutter screen jo WebView editor chalati hai.
- assignment_draft_service.dart — draft save/load (aur purane Quill drafts ko HTML mein badalna).
- html_pdf_printer.dart — Dart se native PDF banwane ka bridge.
- MainActivity.kt + PdfPrint.java — Android native PDF renderer.

Static analysis (flutter analyze) bilkul saaf hai — koi compile error ya warning nahi. Lekin logic aur behaviour ke level par kuch ahem bugs hain jo neech tafseel se likhe gaye hain. Har bug ke saath: severity (kitna ahem), file, masla kya hai, user ke saath kya hota hai, aur hal kya hai.

2. Tool kaise kaam karta hai (chhota khulasa)

- Ek hi "AssignmentGeometry" class poori naap-tol ko control karti hai. Yehi cheez editor aur PDF dono ko di jaati hai, is liye screen par jo dikhta hai aur PDF mein jo aata hai — dono ek jaise hote hain.
- Editor ek lambi safaid sheet hai jo content ke saath barhti rehti hai. Page break asal mein sirf "guide line" hain (dashed line + "N صفحه" label) — content ko todte nahi.
- PDF export wahi HTML browser (WebView) se print kara ke banta hai. Pehle "vector" (asli, select-able text) try hota hai; agar woh fail/timeout ho to "raster" (tasveer) fallback chalta hai.

3. Bugs — Ek Nazar Mein

#	Bug	Severity	Area
1	Raster fallback wala PDF kharab banta hai (margins gayab + line kati hui)	HIGH	PDF / Native
2	Editor guide aur PDF break match nahi karte (table/image par)	HIGH	Pagination
3	Undo theek se kaam nahi karta (table/paste/size par)	MEDIUM	Editor
4	Markdown link mein 'javascript:' URL guzar jaata hai	MEDIUM	Security
5	Paste par tasveerein khamoshi se delete ho jaati hain	MEDIUM	Doc model

#	Bug	Severity	Area
6	Heading padding paginator ko nazar nahi aati	LOW	Pagination
7	App 'inactive' hote hi bar bar save karta hai	LOW	Editor
8	Har open par ~3.5 MB fonts bridge se bhej diye jaate hain	LOW	Performance

4. Bugs — Tafseel Ke Saath

Bug 1 [HIGH] — Raster fallback wala PDF kharab banta hai (margins gayab + line beech se kati)

File: MainActivity.kt → renderRaster(); aur html_pdf_printer.dart

Masla kya hai:

PDF banane ke do tareeqay hain. "Vector" tareeqa theek kaam karta hai. Lekin jab vector fail ya timeout ho jaye (aksar sasti / kam-RAM wali phones par), to "raster" fallback chalta hai.

Ye raster wala code abhi bhi purane page-system ke hisaab se bana hua hai. Naye system mein content ke upar/neeche ke margins sirf "@page" CSS rule se aate hain — aur ye rule sirf vector printing mein lagta hai, raster mein nahi.

Upar se, raster har page ko poori "sheetHeightPx" (margin samet) ki height par kaat deta hai, halaanki asal content bina vertical margin ke continuous chal raha hota hai. Nateeja: har page ke upar/neeche margin gayab, aur text ki line page ke kinare par beech se kat jaati hai.

Iske ilawa:

html_pdf_printer.dart "marginPx" value native side ko bhejta hai, lekin MainActivity.kt usay parhta hi nahi. To raster ke paas margin wapas lagane ka koi zariya bhi mojood nahi.

User ke saath kya hota hai:

Sasti phone wale student ka PDF kharab dikhega — lines aadhi kati hui, aur margins na ke barabar. Aur yehi woh students hain jo is feature par sab se zyada depend karte hain.

Hal (Fix):

- renderRaster ko vector wali geometry ke mutabiq dobara likhein: har page par safaid sheet banayein, content ko contentHeightPx (na ke sheetHeightPx) ke hisaab se shift karein, aur marginPx ka inner margin chhorein.
- Kotlin handler mein marginPx ko parh kar istemal karein (abhi ignore ho raha hai).

Bug 2 [HIGH] — Editor ki guide line aur PDF ka page-break match nahi karte (table / image par)

File: assignment_html_service.dart → pageBreaks() aur export CSS (break-inside: avoid)

Masla kya hai:

Editor ki page-break guide line kisi bhi jagah kat sakti hai — table ya image ke beech mein bhi.

Lekin PDF mein rule hai: "tr, img { break-inside: avoid }" yaani table ki row ya image ko beech se mat todo, poora agle page par bhej do. Is liye screen par guide line table ke beech dikhti hai, jabke PDF us poori row/image ko agle page par le jaata hai.

User ke saath kya hota hai:

Wahi purana masla wapas aa jaata hai — "screen par kuch, PDF mein kuch aur". Khaas tor par jab table ya tasveer page ke kinare par ho. Aur is poore tool ko WYSIWYG (jo dikhe wahi chhpe) banane ka maqsad isi se toot-ta hai.

Hal (Fix):

- pageBreaks() mein jab koi break ek table-row / image / list-item ke andar aaye, to us break ko us block ke top par shift kar dein (PDF ke break-inside: avoid ki naqal). Phir guide aur PDF dono match karenge.

Bug 3 [MEDIUM] — Undo (wapas karna) theek se kaam nahi karta

File: assignment_html_service.dart (cmd vs custom operations)

Masla kya hai:

Bold / italic / align jaise commands "execCommand" se hote hain, jo browser ke undo stack mein aate hain — inka undo theek chalta hai.

Lekin table insert, font-size change, RTL/LTR direction, indent, aur paste — ye sab seedha DOM ko badalte hain aur undo stack mein nahi aate. Aur do alag tarah ke changes mix hone se undo history kharab bhi ho sakti hai.

User ke saath kya hota hai:

Student table laga kar ya kuch paste kar ke Ctrl+Z dabaye to woh wapas nahi hota — ya phir ghalat cheez undo ho jaati hai. Likhne wale tool mein ye roz ki takleef hai.

Hal (Fix):

- Custom operations ke liye apna chhota undo stack banayein: har operation se pehle page ka innerHTML + selection save karein.
- Ya jahan WebView support kare, insertion ko execCommand('insertHTML') se karein taake woh native undo stack mein aaye.

Bug 4 [MEDIUM] — Markdown link mein "javascript:" jaisa khatarnak URL guzar jaata hai

File: assignment_html_service.dart → inlineMd()

Masla kya hai:

Baaki har jagah link ko sirf https / mailto tak mehdood kiya gaya hai (applyLink aur paste cleaner dono).

Lekin jab Markdown text paste hota hai (jaise kisi AI app se), to "inlineMd" wala converter URL ko check nahi karta. To "[x](javascript:...)" ek live link ban jaata hai.

User ke saath kya hota hai:

Filhaal khatra kam hai (editor mein click se navigate nahi hota, aur PDF static hota hai), lekin ye ek security loophole hai aur baaki code ke standard se mukhtalif hai — isay band karna behtar hai.

Hal (Fix):

- inlineMd mein bhi URL ko usi https/mailto allowlist se check karein; agar scheme ghalat ho to sirf text rakhein, link na banayein.

Bug 5 [MEDIUM] — Paste karte waqt tasveerein khamoshi se delete ho jaati hain

File: assignment_html_service.dart → DROP list (paste cleaner)

Masla kya hai:

Paste cleaner ki "DROP" list mein IMG aur SVG shaamil hain, yaani koi bhi paste ki gayi tasveer / diagram bina bataye ghaayab ho jaati hai.

Aur image insert ka koi button bhi nahi hai — halaanki CSS mein "img { max-width:100% }" pehle se mojood hai.

User ke saath kya hota hai:

Student koi diagram, screenshot ya equation ki tasveer paste kare to woh chali jaati hai, aur koi message bhi nahi aata. Ye content ka seedha nuqsan hai.

Hal (Fix):

- Image support add karein: ek insert button (file pick → chhoti ki hui base64 image), aur paste mein image ko drop na karein.
- Export pehle se base64 image ko handle kar leta hai, is liye PDF mein bhi aa jaayegi.

Bug 6 [LOW] — Heading ki padding paginator ko nazar nahi aati (guide thori khisak jaati hai)

File: assignment_html_service.dart → _blockCss / pageBreaks

Masla kya hai:

Page break ka hisaab text ki line ke "client rects" se hota hai. Lekin heading apni jagah "padding" se banati hai, aur range ke rects mein padding shaamil nahi hoti.

Is liye heading ke baad guide line asal flow se thori upar/neeche aa sakti hai. Code ka comment purane (offsetHeight wale) system ki baat karta hai jo ab istemal hi nahi hota.

User ke saath kya hota hai:

Sirf thora sa visual farq guide line mein. Bara masla nahi, lekin theek karne layak hai.

Hal (Fix):

- Block ke neeche ki padding ko hisaab mein lein, ya kam az kam purana comment update karein taake aglay banday ko confusion na ho.

Bug 7 [LOW] — App "inactive" hote hi bar bar save karta hai

File: web_assignment_editor_screen.dart → didChangeAppLifecycleState

Masla kya hai:

"paused" ke saath "inactive" state par bhi save hota hai. Lekin "inactive" to bohot baar trigger hota hai — dialog khulne par, notification shade kheenchne par, permission prompt par — aur har baar poora `getBody()` + file write chalta hai.

User ke saath kya hota hai:

Faaltu kaam (battery / performance thoda zaaya). Masla bada nahi.

Hal (Fix):

- Sirf "paused" par save karein — yehi normal tareeqa hai "app background mein chala gaya" ke liye.

Bug 8 [LOW] — Har baar editor khulte hi ~3.5 MB fonts bridge se bhej diye jaate hain

File: `web_assignment_editor_screen.dart` → `_preloadFonts`

Masla kya hai:

Editor load hote hi saare fonts inject ho jaate hain (akela Noto Nastaliq ~1.2 MB), base64 mein 3 alag JavaScript calls se.

Ye "lazy injection" wale maqsad ke khilaaf hai (jiska comment code mein likha bhi hai), aur kam-power phones par shuru mein lag (jank) la sakta hai.

User ke saath kya hota hai:

Editor khulne mein sasti phone par thori dair / atak. Bara masla nahi.

Hal (Fix):

- Sirf woh fonts preload karein jo draft mein asal mein use ho rahe hain; baaki ko zaroorat parne par inject karein (`_ensureFont` ka raasta pehle se mojud hai).

5. Tajaweez (Behtari ke Mashvaray)

Layout

- Manual page break (Ctrl+Enter): student naya section / baab naye page se shuru kar sake.
- Accessibility: editor mein zoom band hai (`user-scalable=no`). Kam-nazar users ke liye zoom allow karein — page geometry par koi farq nahi parega.

Render

- Image support (Bug 5 ke saath): insert button + paste mein image rakhein. Export pehle hi base64 images support karta hai.
- Table behtar banayein: column width drag, aur cell merge/split. Abhi sab columns barabar hain aur resize nahi hote.
- Widow/orphan control: PDF mein kabhi ek akeli line stranded ho sakti hai; behtar setting di ja sakti hai (page bharne aur khoobsurti mein balance).

Document model

- Save par normalize karein: model abhi raw innerHTML hai; save par halka sa sanitize karein taake hamesha theek blocks hon aur PDF predictable rahe.
- Stored HTML ko version dein (schemaVersion): future mein format migration mehfooz tareeqe se ho sake.
- Dead code hatayein: cleanClone() abhi .tail/.gap/.brk strip karta hai jo naye continuous editor mein bante hi nahi.

6. Pehle Kya Theek Karein (Priority)

1. Bug 1 (raster fallback) — asli users ko ghalat PDF milta hai. Sab se pehle yehi.
2. Bug 2 (table/image: guide vs PDF) — WYSIWYG (jo dikhe wahi chhpe) ka wada toot-ta hai.
3. Bug 3 (undo) — rozmarra ke likhne mein sab se bari takleef.
4. Phir Bug 4 + Bug 5 (link safety + image drop) — jaldi ho jaane wali hardening.

Note: Agar aap kahein to main inme se koi bhi bug khud theek kar sakta hoon. Bata dein kaunsa pehle karoon — main mashwara dunga ke Bug 1 aur Bug 2 saath theek karein (dono pagination-geometry ke hain aur ek hi export se test ho jaate hain).